

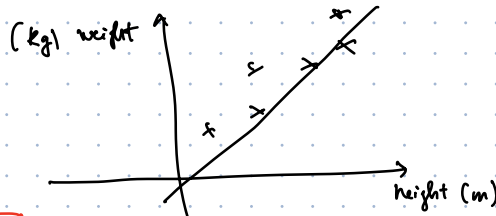
Exercise 1

Regression

$$f: X \rightarrow Y \subset \mathbb{R}$$

$$p(x, y) = C e^{-x^2} e^{-(x-y)^2}$$

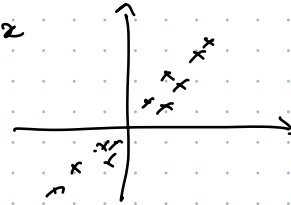
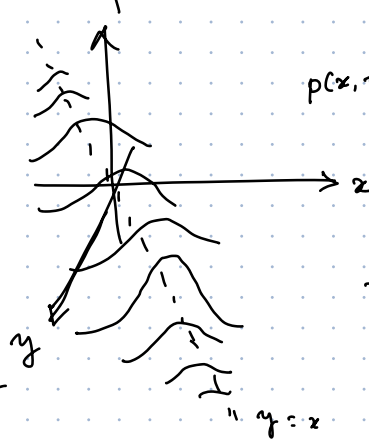
This is a pdf defined on $\mathbb{R}^2$.



$$p(x, y) = p(x) \cdot p(y|x)$$

1. Plot $p(x, y)$

$$L(f) = \int_{\mathbb{R}^2} \ell(f(x), y) p(x, y) dx dy$$

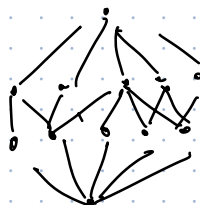
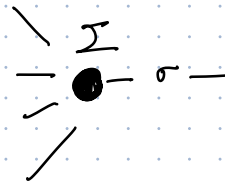
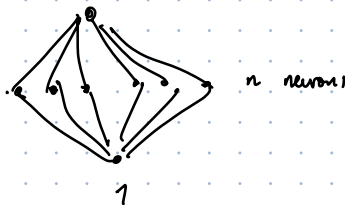


We want to train a model (FNN) on a Dataset (training set) that comes from $p(x, y)$

2. Sample a Training Set $\underline{D} = \{(x_1, y_1), \dots, (x_N, y_N)\}$ from $p(x, y)$

3. Use this training set to train an MLP (multi layer perceptron)

$$\ell(a, b) = \frac{1}{2} (a - b)^2$$



In order to solve 3

- Import data that you have generated in step 1
- Define the NN (using pytorch libraries)
- Define the training loss
- Write the training loop

C is the number such that

$$\int_{\mathbb{R}^2} C e^{-x^2} e^{-(x-y)^2} dx dy = 1$$

$$C = \frac{1}{\int_{\mathbb{R}^2} e^{-x^2} e^{-(x-y)^2} dx dy}$$

$$\int_{-\infty}^{+\infty} e^{-x^2} dx = \sqrt{\pi}$$



Exercise 2: Classification problem with FNN

0. If you haven't done MNIST (pytorch has this dataset)

- * Choose the loss function
- Choose the optimizer
- Do training loop
- Compute the accuracy on test set.

1. Do the same thing with FMNIST

$$(batch_size, \#columns, W, H)$$



IRIS is another dataset for classifications of flowers